

A COUNTEREXAMPLE TO JOHNSTON AND RUSSO'S UNAMBIGUOUS PPT-DISCRIMINATION CONJECTURE

ABSTRACT. Johnston and Russo conjectured that every orthonormal locally diagonal orthogonally invariant basis is unambiguously distinguishable by positive-partial-transpose measurements. We determine the optimal zero-error PPT success probability of an arbitrary complete orthonormal bipartite basis: it equals the total prior weight of its product vectors. Hence such a basis is unambiguously PPT-distinguishable, with positive conclusive probability for every basis state, if and only if it is a product basis. The two-qubit Bell basis, which Johnston and Russo identify as an orthonormal LDOI basis, therefore refutes their conjecture as stated in arXiv:2604.12808v1.

1. COMPLETE ORTHONORMAL BASES

Johnston and Russo state [1, Conjecture 1] as follows.

Every orthonormal LDOI basis of $\mathcal{X} \otimes \mathcal{Y}$ can be unambiguously distinguished by PPT measurements.

Here LDOI abbreviates locally diagonal orthogonally invariant, and PPT positive partial transpose. Johnston and Russo do not define unambiguous distinguishability, so we adopt the definition of [2, Sec. II]: a collection $\rho_1, \dots, \rho_N$ is unambiguously distinguishable by PPT measurements if there is a PPT POVM $(\Pi_k)_{k=0}^N$ with $\text{tr}(\Pi_k \rho_j) = p_k \delta_{kj}$ for $1 \leq k, j \leq N$ and $p_k > 0$ for every k , the outcome $k = 0$ being the inconclusive one. That this is the intended reading is settled by the source itself, which states that a proof of the conjecture “would require constructing a nontrivial PPT POVM achieving unambiguous discrimination” [1]: under the weaker reading in which the conclusive probabilities are unconstrained, the all-inconclusive POVM already witnesses the conjecture for every basis (Remark 5).

The analogue of the criterion below for perfect discrimination is already known. Yu, Duan and Ying record that “an orthonormal basis of $\mathcal{X} \otimes \mathcal{Y}$ is perfectly distinguishable by PPT POVMs if and only if it is a product basis” [2, Sec. II], and the remarks following [1, Theorem 10] record that “an orthonormal LDOI basis is perfectly distinguishable by PPT measurements if and only if it consists entirely of product states”. Theorem 1 is an immediate consequence of the same completeness reduction, which forces every conclusive outcome to be a multiple of a basis projector. What we

2020 *Mathematics Subject Classification*. Primary 81P15; Secondary 81P45, 15A69.

Key words and phrases. PPT discrimination, unambiguous discrimination, entanglement, counterexample.

add is Corollary 3, which evaluates the resulting quantity on the LDOI parameterization of [1, Definition 6], and the observation that feasibility of the associated semidefinite program does not certify unambiguous discrimination (Remark 5).

Let $\mathcal{H} = \mathcal{X} \otimes \mathcal{Y}$ be finite-dimensional, let Γ denote partial transpose on $\mathcal{X}$, and let $\mathcal{B} = \{|\psi_1\rangle, \dots, |\psi_D\rangle\}$ be an orthonormal basis of $\mathcal{H}$. For positive priors $p_1, \dots, p_D$ summing to one, define the optimal zero-error PPT success probability by

$$(1) \quad \omega_0^{\text{PPT}}(\mathcal{B}, p) = \max \sum_{i=1}^D p_i \langle \psi_i | M_i | \psi_i \rangle,$$

where the maximum is over POVMs $\{M_1, \dots, M_D, M_?\}$ such that every outcome is PPT and

$$\langle \psi_j | M_i | \psi_j \rangle = 0 \quad (i \neq j).$$

Theorem 1. *For every complete orthonormal bipartite basis,*

$$\omega_0^{\text{PPT}}(\mathcal{B}, p) = \sum_{\substack{1 \leq i \leq D \\ |\psi_i\rangle \text{ is product}}} p_i.$$

Proof. Fix a feasible POVM. Since $M_i \succeq 0$, the zero-error conditions imply

$$0 = \langle \psi_j | M_i | \psi_j \rangle = \|M_i^{1/2} |\psi_j\rangle\|^2,$$

so $M_i |\psi_j\rangle = 0$ whenever $j \neq i$. Completeness of the basis and Hermiticity of M_i give

$$M_i = c_i |\psi_i\rangle\langle\psi_i|.$$

Since $0 \preceq M_i \preceq I$, we have $0 \leq c_i \leq 1$. If $|\psi_i\rangle$ is entangled, write a Schmidt decomposition

$$|\psi_i\rangle = \sum_{k=1}^r s_k |x_k\rangle \otimes |y_k\rangle, \quad r \geq 2,$$

with $s_k > 0$. In Schmidt bases, the partial transpose of $|\psi_i\rangle\langle\psi_i|$ has eigenvalues s_k^2 and $\pm s_k s_\ell$ for $k < \ell$, together with possible zero eigenvalues. It is therefore not positive semidefinite. Since $M_i^\Gamma \succeq 0$, this forces $c_i = 0$. Thus the objective in (1) is at most the stated sum. Only the PPT condition on the conclusive outcomes $M_1, \dots, M_D$ was used here, so the bound holds even if $M_?$ is not required to be PPT.

For the reverse inequality, let S be the set of indices for which $|\psi_i\rangle = |x_i\rangle \otimes |y_i\rangle$ is product, and set

$$M_i = \begin{cases} |\psi_i\rangle\langle\psi_i|, & i \in S, \\ 0, & i \notin S, \end{cases} \quad M_? = I - \sum_{i \in S} |\psi_i\rangle\langle\psi_i|.$$

The vectors $|\bar{x}_i\rangle \otimes |y_i\rangle$, $i \in S$, are orthonormal: for $i \neq j$, orthogonality of the original product vectors gives $\langle x_i, x_j \rangle \langle y_i, y_j \rangle = 0$, and hence also

$\overline{\langle x_i, x_j \rangle} \langle y_i, y_j \rangle = 0$. Therefore

$$M_?^\Gamma = I - \sum_{i \in S} |\bar{x}_i \otimes y_i\rangle \langle \bar{x}_i \otimes y_i| \succeq 0.$$

All other outcomes are also PPT, so this POVM attains the stated value. $\square$

Corollary 2. *A complete orthonormal bipartite basis is unambiguously distinguishable by a PPT measurement, with positive conclusive probability for every basis state, if and only if every basis vector is product.*

2. LDOI BASES AND THE COUNTEREXAMPLE

We recall the parameterization in [1, Definition 6]. Let $U = (u_{ik}) \in \mathbf{U}(n)$ and let $A = (a_{ij}) \in \mathbf{L}(\mathbb{C}^n)$ satisfy

$$|a_{ij}|^2 + |a_{ji}|^2 = 1 \quad (i < j).$$

The corresponding orthonormal LDOI basis consists of

$$\begin{aligned} |\phi_{ii}\rangle &= \sum_{k=1}^n u_{ik} |kk\rangle, \\ |\phi_{ij}\rangle &= a_{ij} |ij\rangle + \bar{a}_{ji} |ji\rangle, \\ |\phi_{ji}\rangle &= a_{ji} |ij\rangle - \bar{a}_{ij} |ji\rangle \quad (i < j). \end{aligned}$$

Define

$$\begin{aligned} r(U) &= \#\{i : \text{row } i \text{ of } U \text{ has exactly one nonzero entry}\}, \\ z(A) &= \#\{(i, j) : i < j, a_{ij}a_{ji} = 0\}. \end{aligned}$$

A unitary matrix is called *monomial* if it has exactly one nonzero entry in each row and column.

Corollary 3. *For the uniform ensemble on the LDOI basis parameterized by (U, A) ,*

$$\omega_0^{\text{PPT}}(U, A) = \frac{r(U) + 2z(A)}{n^2}.$$

In particular, the basis is unambiguously PPT-distinguishable in the standard sense if and only if U is monomial and

$$a_{ij}a_{ji} = 0 \quad (i < j).$$

Proof. The coefficient matrix of $|\phi_{ii}\rangle$ is diagonal with entries $u_{i1}, \dots, u_{in}$, so this vector is product exactly when row i of U has one nonzero entry. For $i < j$, the coefficient matrix of $|\phi_{ij}\rangle$ (respectively $|\phi_{ji}\rangle$) is supported on the rows and columns indexed by $\{i, j\}$, so its rank equals the rank of its 2×2 submatrix there, whose determinant is $-a_{ij}\bar{a}_{ji}$ (respectively $a_{ji}\bar{a}_{ij}$). Since $|a_{ij}|^2 + |a_{ji}|^2 = 1$, that submatrix is nonzero, hence has rank 1 exactly when its determinant vanishes. Thus both vectors are product exactly when $a_{ij}a_{ji} = 0$. There are therefore $r(U) + 2z(A)$ product vectors, and the formula follows from Theorem 1. If every row of a unitary matrix has one nonzero entry, then the matrix is monomial. $\square$

Corollary 4. *Conjecture 1 of [1], as stated in arXiv:2604.12808v1, is false in local dimension $n = 2$.*

Proof. Johnston and Russo’s Example 8 takes

$$U = A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

and obtains the Bell basis

$$\frac{|11\rangle \pm |22\rangle}{\sqrt{2}}, \quad \frac{|12\rangle \pm |21\rangle}{\sqrt{2}}.$$

Here $r(U) = z(A) = 0$, so Corollary 3 gives $\omega_0^{\text{PPT}} = 0$. In particular, no Bell state has a conclusive PPT outcome of positive probability. $\square$

Remark 5 (Feasibility). The all-inconclusive POVM $M_i = 0$ and $M_? = I$ is always feasible under the PPT, completeness, and zero-error constraints. Consequently, feasibility alone does not certify unambiguous discrimination. The evidence offered for Conjecture 1 is numerical: Johnston and Russo report that it “has been verified numerically for random LDOI bases in dimensions $n = 2, 3, 4$, and 5” and that “the SDP for unambiguous PPT discrimination was feasible for every randomly generated LDOI basis tested” [1]. Their implementation, reported there as part of the `toqito` package, we have not inspected; if the feasibility test imposed no lower bound on the conclusive probabilities, then it is met by $M_i = 0$, $M_? = I$ for every basis and so carries no information about the conjecture. One may instead maximize a scalar t subject to

$$\langle \psi_i | M_i | \psi_i \rangle \geq t \quad (1 \leq i \leq D);$$

the standard unambiguous-discrimination condition holds exactly when the optimal value is positive.

REFERENCES

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